Haloalkanes and Haloarenes

> *Halogen compounds hold out in the environment, because soil bacteria break them down only slowly.*

By the end of this unit

- read a structure and give the haloalkane or haloarene its IUPAC name
- set out the reactions that make haloalkanes and haloarenes, and the reactions they go on to give
- tie the structure of each family to the kinds of reaction it enters
- use stereochemistry to read a reaction mechanism
- see what organo-metallic compounds are good for
- weigh what polyhalogen compounds do to the environment

Where halogen compounds turn up

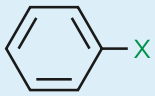
- **Solvents** — for compounds that are relatively non-polar
- **Starting materials** — for a wide range of organic syntheses
- **Chloramphenicol** — chlorine-containing antibiotic made by microorganisms; effective against typhoid fever
- **Thyroxine** — the body's own iodine-containing hormone; too little of it causes goiter
- **Chloroquine** — synthetic, used against malaria
- **Halothane** — anaesthetic used during surgery
- **Fully fluorinated compounds** — under consideration as blood substitutes in surgery

The two families

- Replace one or more hydrogen atoms of an **aliphatic** hydrocarbon by halogen and the product is an **alkyl halide**, or haloalkane; do the same to an **aromatic** hydrocarbon and it is an **aryl halide**, or haloarene.
- The difference sits at the carbon: in a haloalkane the halogen is on an sp^3 carbon of an alkyl group, in a haloarene on an sp^2 carbon of an aryl group.

R—X

Haloalkane · sp^3 C–X



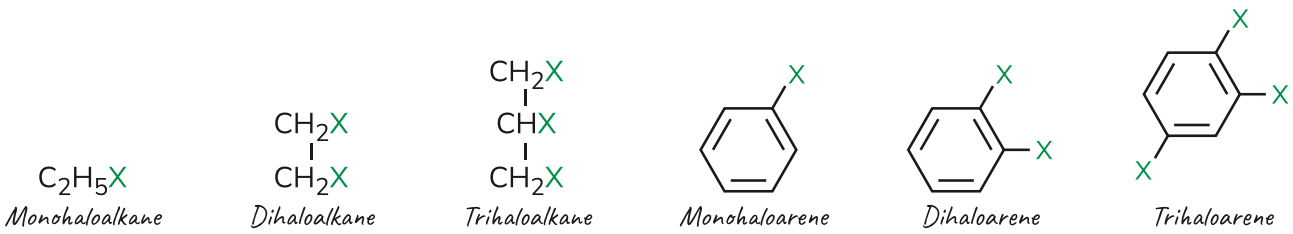
Haloarene · sp^2 C–X

The unit runs in two parts: **I · Haloalkanes and haloarenes**, then **II · Polyhalogen compounds**.

A.1 Classification

First count: how many halogen atoms

- One halogen atom makes a **monohalogen** compound, two a **dihalogen** compound, and three or more a **polyhalogen** compound (tri-, tetra- and so on).



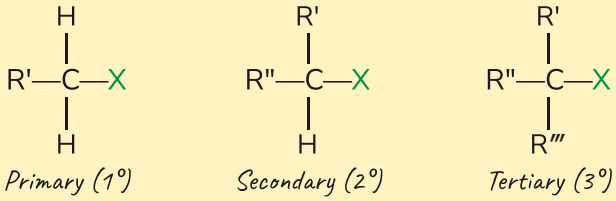
Then sort the monohalogen ones by the carbon that holds the halogen

- the dividing question is how that carbon is **hybridised** — sp^3 in one class, sp^2 in the other

Compounds with an sp^3 C–X bond (X = F, Cl, Br, I)

(a) Alkyl halides, or haloalkanes (R–X)

- the halogen sits on an **alkyl group** (R), and the family forms a homologous series $\mathbf{C_nH_{2n+1}X}$
- the carbon carrying the halogen decides the label: on a primary carbon it is a **primary (1°)** alkyl halide, on a secondary carbon a **secondary (2°)**, on a tertiary carbon a **tertiary (3°)**



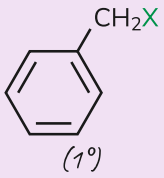
(b) Allylic halides

- the halogen is on an sp^3 carbon that stands **next to a C=C double bond** — an **allylic carbon**

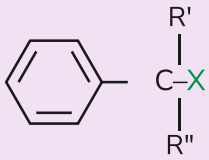
(c) Benzylic halides

- the halogen is on an sp^3 carbon **attached to an aromatic ring**

(c) Benzylic halides · what they look like



(1°)



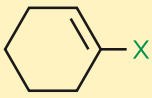
R′ = CH_3, R″ = H (2°) · R′ = R″ = CH_3 (3°)

Compounds with an sp^2 C–X bond

(a) Vinylic halides

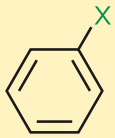
- the halogen is on an sp^2 carbon that belongs to a **carbon–carbon double bond** (C = C)

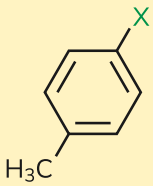
CH_2=CH–X



(b) Aryl halides

- the halogen is bonded straight to an sp^2 carbon **of an aromatic ring**

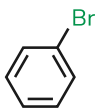
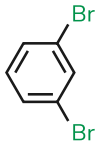
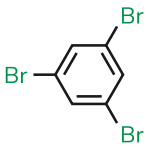




A.2 Nomenclature

- **Common name** of an alkyl halide: name the alkyl group, then the halide.
- **IUPAC name**: treat it as a **halosubstituted hydrocarbon**.
- For a benzene ring carrying **one** halogen the two systems give the **same** name.
- For **two** halogens the common system uses the prefixes ***o-***, ***m-***, ***p-***, while IUPAC uses the locants **1,2 · 1,3 · 1,4**.

Structure	Common name · IUPAC name
$CH_3CH_2CH_2Br$	*n*-Propyl bromide 1-Bromopropane
$H_3C–CH(Cl)–CH_3$	Isopropyl chloride 2-Chloropropane
$H_3C–CH(CH_3)–CH_2Cl$	Isobutyl chloride 1-Chloro-2-methylpropane
$H_3C–C(CH_3)_2–CH_2Cl$	1-Chloro-2,2-dimethylpropane

Structure	Common name · IUPAC name
	Bromobenzene Bromobenzene
	*m*-Dibromobenzene 1,3-Dibromobenzene
	*sym*-Tribromobenzene 1,3,5-Tribromobenzene
$H_3C–CH(Br)–CH_3$	2-Bromopropane

Two halogens of the same kind

- such dihaloalkanes are named as **alkylidene** or **alkylene dihalides**
- both halogens on the **same** carbon — **geminal** halides, or *gem*-dihalides

Both halogens on neighbouring carbons

- these are **vicinal** halides, or *vic*-dihalides
- common names: a *gem*-dihalide is an **alkylidene halide**, a *vic*-dihalide an **alkylene dihalide**
- IUPAC names both of them as **dihaloalkanes**

$H_3C–CHCl_2$

Ethylidene chloride (gem)

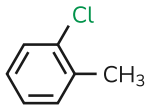
1, 1-Dichloroethane

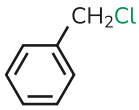
$H_2C(Cl)–CH_2Cl$

Ethylene dichloride (vic)

1, 2-Dichloroethane

Table 6.A **Common and IUPAC Names of some Halides**

Structure	Common name · IUPAC name
$CH_3CH_2CH(Cl)CH_3$	*sec*-Butyl chloride 2-Chlorobutane
$(CH_3)_3CCH_2Br$	*neo*-Pentyl bromide 1-Bromo-2,2-dimethylpropane
$(CH_3)_3CBr$	*tert*-Butyl bromide 2-Bromo-2-methylpropane
$CH_2 = CHCl$	Vinyl chloride Chloroethene
$CH_2 = CHCH_2Br$	Allyl bromide 3-Bromopropene
	*o*-Chlorotoluene 1-Chloro-2-methylbenzene *or* 2-Chlorotoluene

Structure	Common name · IUPAC name
	Benzyl chloride Chlorophenylmethane
CH_2Cl_2	Methylene chloride Dichloromethane
$CHCl_3$	Chloroform Trichloromethane
$CHBr_3$	Bromoform Tribromomethane
CCl_4	Carbon tetrachloride Tetrachloromethane
$CH_3CH_2CH_2F$	*n*-Propyl fluoride 1-Fluoropropane

Worked example 6.a

Draw the structures of all eight structural isomers of molecular formula $C_5H_{11}Br$. Give each its IUPAC name and say whether it is a primary, secondary or tertiary bromide.

$CH_3CH_2CH_2CH_2CH_2Br$ 1-Bromopentane (1°)

$CH_3CH_2CH_2CH(Br)CH_3$ 2-Bromopentane (2°)

$CH_3CH_2CH(Br)CH_2CH_3$ 3-Bromopentane (2°)

$(CH_3)_2CHCH_2CH_2Br$ 1-Bromo-3-methylbutane (1°)

The remaining four isomers are on sheet 5.

Worked example 6.a · *the last four isomers*

$(CH_3)_2CHCHBrCH_3$ 2-Bromo-3-methylbutane (2°)

$(CH_3)_2CBrCH_2CH_3$ 2-Bromo-2-methylbutane (3°)

$CH_3CH_2CH(CH_3)CH_2Br$ 1-Bromo-2-methylbutane (1°)

$(CH_3)_3CCH_2Br$ 1-Bromo-2,2-dimethylpropane (1°)

Worked example 6.b

Write the IUPAC name of each structure below.

(i) $CH_3–CH=CH–CHBr–CH_3$
4-Bromopent-2-ene

(ii) $CH_2=C(CH_3)–CHBr–CH_3$
3-Bromo-2-methylbut-1-ene

(iii) $CH_3–CH=C(CH_3)–CHBr–CH_3$
4-Bromo-3-methylpent-2-ene

(iv) $BrCH_2–C(CH_3)=CH–CH_3$
1-Bromo-2-methylbut-2-ene

(v) $BrCH_2–CH=CH–CH_3$
1-Bromobut-2-ene

(vi) $CH_2=C(CH_3)–CH_2Br$
3-Bromo-2-methylpropene

Intext question

Q1 Write structures for: (i) 2-Chloro-3-methylpentane; (ii) 1-Chloro-4-ethylcyclohexane; (iii) 4-*tert*. Butyl-3-iodoheptane; (iv) 1,4-Dibromobut-2-ene; (v) 1-Bromo-4-*sec*. butyl-2-methylbenzene.

A.3 Nature of the C–X bond

- halogen is **more electronegative than carbon**, so the carbon–halogen bond of an alkyl halide is **polarised**
- the carbon carries a **partial positive** charge, the halogen a **partial negative** one
- going **down the group** the halogen atom grows — fluorine smallest, iodine largest — so the C—X **bond length rises from C—F to C—I**
- Table 6.B collects, for each C—X bond, its length, its enthalpy and its dipole moment

—Cδ+—Xδ–

- alkyl halides are best made from **alcohols**, which are easy to come by

Table 6.B **Carbon-Halogen (C–X) Bond Lengths, Bond Enthalpies and Dipole Moments**

Bond	Bond length/pm	C-X Bond enthalpies/ kJmol^{-1}	Dipole moment/Debye
CH_3–F	139	452	1.847
CH_3–Cl	178	351	1.860
CH_3–Br	193	293	1.830
CH_3–I	214	234	1.636

A.4 Making haloalkanes · from alcohols

- halogen takes the place of an alcohol's **hydroxyl group** when it meets a **halogen acid of high concentration**, a **phosphorus halide**, or **thionyl chloride**
- **thionyl chloride is the one to choose**: SO_2 and HCl leave as gases, so what remains is pure alkyl halide
- 1° and 2° alcohols will not take HCl without the catalyst **$ZnCl_2$**; a 3° alcohol asks no more than a shake at room temperature with HCl of high concentration, and **constant boiling HBr (48%)** serves for the bromides
- heating the alcohol with **NaI or KI in orthophosphoric acid of 95% strength** gives R—I in good yield
- faced with one and the same haloacid, alcohols line up **3° > 2° > 1°**
- PBr_3 and PI_3 are usually made ***in situ*** (produced in the reaction mixture) from red phosphorus with bromine and with iodine

$$R\text{–}OH + HCl \xrightarrow{ZnCl_2} R\text{–}Cl + H_2O$$

$$R\text{–}OH + NaBr + H_2SO_4 \longrightarrow R\text{–}Br + NaHSO_4 + H_2O$$

$$3R\text{–}OH + PX_3 \longrightarrow 3R\text{–}X + H_3PO_3 \quad (X = Cl, Br)$$

$$R\text{–}OH + PCl_5 \longrightarrow R\text{–}Cl + POCl_3 + HCl$$

$$R\text{–}OH \xrightarrow[X_2 = Br_2, I_2]{\text{red } P/X_2} R\text{–}X$$

$$R\text{–}OH + SOCl_2 \longrightarrow R\text{–}Cl + SO_2 + HCl$$

Running the chloride preparation

- either pass **dry hydrogen chloride gas** through a solution of the alcohol, or **heat** the alcohol with concentrated aqueous halogen acid

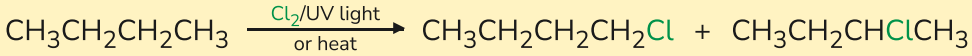
Why none of this makes an aryl halide

- in a phenol the **carbon–oxygen bond has partial double bond character**
- being stronger than a single bond, it is **hard to break**

From hydrocarbons · (I) free radical halogenation of alkanes

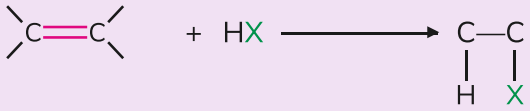
- chlorinating or brominating an alkane by the free radical route leaves a **tangle** of isomeric mono- and polyhaloalkanes, hard to separate — continued on sheet 7

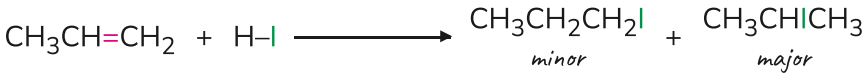
- the isomers are hard to separate as pure compounds, so the **yield of any single one is low**



From alkenes · (i) adding a hydrogen halide

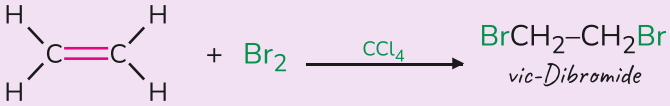
- **HCl, HBr or HI** turns an alkene into the matching alkyl halide
- propene gives **two** products, but only one predominates, as **Markovnikov's rule** says (Unit 13, Class XI)





From alkenes · (ii) adding a halogen

- in the laboratory, **bromine in CCl_4** is added to an alkene and its **reddish brown colour is discharged**; that discharge is an important way to **detect a double bond** in a molecule
- the addition yields colourless ***vic*-dibromides** (Unit 9, Class XI)



Worked example 6.c

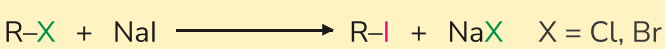
Identify every monochloro structural isomer expected from free radical monochlorination of $(CH_3)_2CHCH_2CH_3$.

The molecule holds **four different kinds of hydrogen atom**. Replacing each in turn gives:

$(CH_3)_2CHCH_2CH_2Cl$
$(CH_3)_2CHCH(Cl)CH_3$
$(CH_3)_2C(Cl)CH_2CH_3$
$CH_3CH(CH_2Cl)CH_2CH_3$

Halogen exchange

- alkyl iodides are often made from alkyl **chlorides or bromides with NaI in dry acetone** — the **Finkelstein reaction**
- the NaCl or NaBr formed is **precipitated in dry acetone**, which pushes the forward reaction along by **Le Chatelier's Principle**
- alkyl fluorides are best made by heating an alkyl chloride or bromide with a **metallic fluoride** — continued on sheet 8



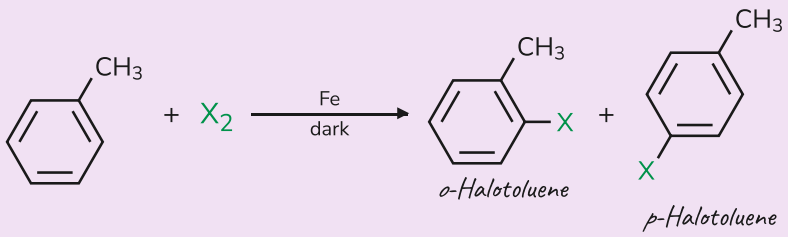
- the metallic fluoride may be **AgF, Hg_2F_2, CoF_2 or SbF_3**; the reaction is called the **Swarts reaction**

$$H_3C{-}Br + AgF \longrightarrow H_3C{-}F + AgBr$$

A.5 Making haloarenes

(i) From hydrocarbons, by attack of an electrophile

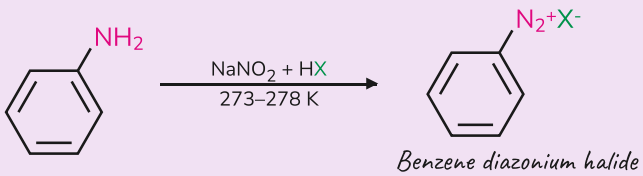
- chlorine and bromine substitute into an arene readily, over a **Lewis acid catalyst** such as iron or iron(III) chloride, and the aryl **chloride or bromide** results

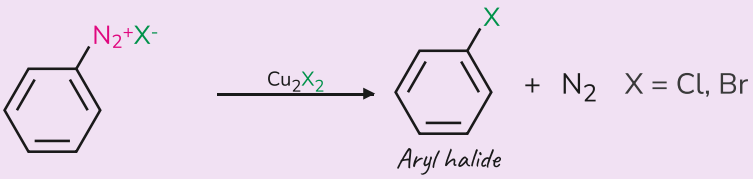


- the ***ortho* and *para* isomers separate easily**, because their melting points differ widely
- reactions with **iodine run both ways**, so an **oxidising agent** (HNO_3, HIO_4) must destroy the HI that iodination throws off
- this route **does not serve for fluoro compounds** — fluorine is far too reactive

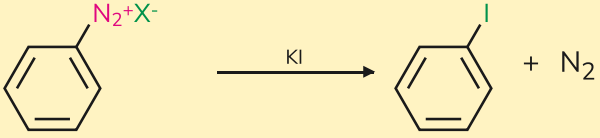
(ii) From amines, by Sandmeyer's reaction

- hold a **primary aromatic amine** in cold aqueous mineral acid, in solution or in suspension, add **sodium nitrite**, and a **diazonium salt** forms
- mixing that freshly prepared solution with **cuprous chloride or cuprous bromide** replaces the diazonium group by **–Cl** or **–Br**



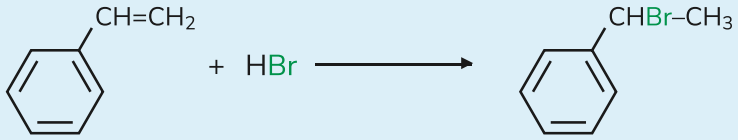


- swapping the diazonium group for **iodine needs no cuprous halide** — simply shake the diazonium salt with **potassium iodide**

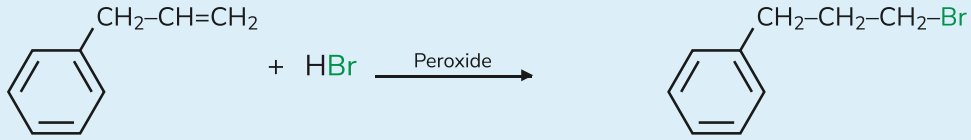


Worked example 6.d

Write the products of these reactions.

(i) 

(ii) $CH_3–CH_2–CH=CH_2 + HCl \longrightarrow CH_3–CH_2–\underset{\displaystyle Cl}{CH}–CH_3$

(iii) 

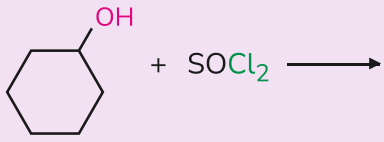
Intext questions

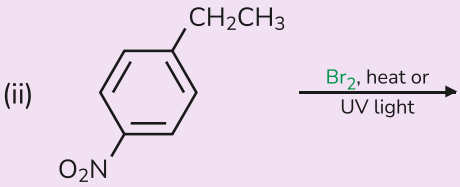
Q2 Why does sulphuric acid have no place when an alcohol is reacted with KI?

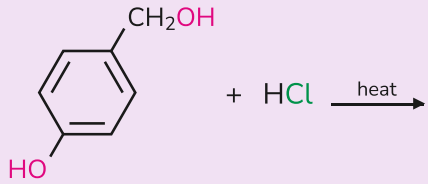
Q3 Write structures of different dihalogen derivatives of propane.

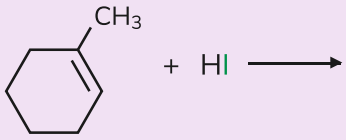
Q4 Of the isomeric alkanes C_5H_{12}, which one gives, under photochemical chlorination, (i) one monochloride only, (ii) three monochloride isomers, (iii) four of them?

Q5 Draw the structures of the major monohalo products of each reaction below.

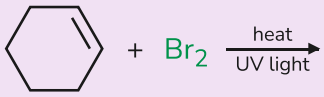
(i) 

(ii) 

(iii) 

(iv) 

(v) $CH_3CH_2Br + NaI \longrightarrow$

(vi) 

A.6 Physical properties

- pure alkyl halides are **colourless**, but bromides and iodides **develop colour on exposure to light**
- many volatile halogen compounds have a **sweet smell**

Melting and boiling points

- **gases at room temperature**: methyl chloride, methyl bromide, ethyl chloride and some chlorofluoromethanes
- **higher members** are liquids or solids
- as a rule the molecules of an organic halogen compound are **polar**

- set against the parent hydrocarbon they carry **greater polarity and higher molecular mass**, which leaves the forces between molecules — **dipole–dipole together with van der Waals** — the stronger
- hence chlorides, bromides and iodides **boil considerably higher** than hydrocarbons of comparable molecular mass

- the attractions strengthen as molecules grow **larger and carry more electrons**
- for one and the same alkyl group the boiling points fall in the order **RI > RBr > RCl > RF**, because van der Waals forces grow with the **size and mass of the halogen atom**

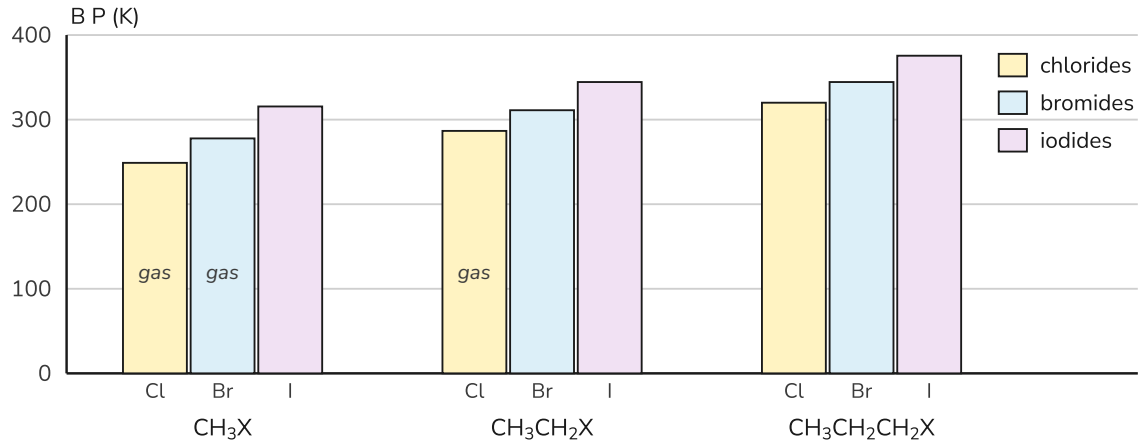
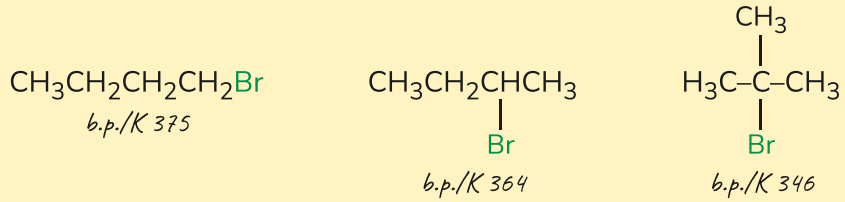
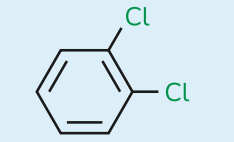


Figure 6.A Comparison of boiling points of some alkyl halides

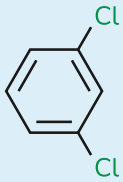
- among isomeric haloalkanes the boiling point **falls as branching increases** — of the three isomers below, 2-bromo-2-methylpropane boils lowest



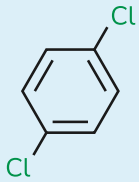
- isomeric dihalobenzenes boil at **very nearly the same** temperature
- the ***para* isomers melt high** compared with the *ortho* and *meta* ones: a *para* isomer is symmetrical and so **fits the crystal lattice better**



b.p/K 453 · m.p/K 256



b.p/K 446 · m.p/K 249



b.p/K 448 · m.p/K 323

Density

- hydrocarbon derivatives carrying **bromine, iodine or several chlorines** all **outweigh water**
- density climbs with the **number of carbon atoms**, the **number of halogen atoms** and the **atomic mass of the halogen** (Table 6.C)

TABLE 6.C **Density of Some Haloalkanes**

Compound	Density (g/mL)	Compound	Density (g/mL)
n–C_3H_7Cl	0.89	CH_2Cl_2	1.336
n–C_3H_7Br	1.335	$CHCl_3$	1.489
n–C_3H_7I	1.747	CCl_4	1.595

Solubility

- haloalkanes are **only very slightly soluble in water**
- dissolving one costs energy: the attractions between haloalkane molecules must be overcome and the **hydrogen bonds between water molecules broken**
- less energy comes back when the new haloalkane–water attractions form, since these are **weaker than the hydrogen bonds** lost
- so the solubility of haloalkanes in water is **low**
- they do dissolve in **organic solvents**, because the new haloalkane–solvent attractions are about **as strong as the ones broken** in the separate haloalkane and solvent

Intext question

Q6 Arrange each set in order of increasing boiling point. (i) Bromomethane, Bromoform, Chloromethane, Dibromomethane. (ii) 1-Chloropropane, Isopropyl chloride, 1-Chlorobutane.

A.7 Chemical reactions of haloalkanes

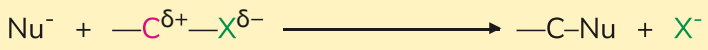
Three categories

- substitution by a **nucleophile**
- **elimination**
- reaction with a **metal**

(1) Nucleophilic substitution

- a **nucleophile is rich in electrons** (Class XI), so its attack falls where the substrate is **poor in them**
- sheet 12 sets out what follows when one nucleophile displaces another

- where an incoming nucleophile **displaces one already held in the molecule**, the reaction is a **nucleophilic substitution**, and the haloalkane is the **substrate**
- the nucleophile attacks the haloalkane at the carbon bonded to halogen, which carries a **partial positive charge**
- substitution follows, and the halogen atom — the **leaving group** — departs as a **halide ion**
- because a nucleophile starts it, the reaction takes that name



- this is **one of the most useful classes of organic reaction** of alkyl halides, in which the halogen sits on an ***sp*³ hybridised carbon**
- the products formed with some common nucleophiles are collected in Table 6.D

Table 6.D Nucleophilic Substitution of Alkyl Halides (R–X)

$$R - X + Nu^- \rightarrow R - Nu + X^-$$

Reagent	Nucleophile (Nu^-)	Substitution product R–Nu	Class of main product
NaOH (KOH)	HO^-	ROH	Alcohol
H_2O	H_2O	ROH	Alcohol
NaOR'	$R'O^-$	ROR'	Ether
NaI	I^-	R—I	Alkyl iodide
NH_3	NH_3	RNH_2	Primary amine
$R'NH_2$	$R'NH_2$	RNHR'	Sec. amine
R'R"NH	R'R"NH	RNR'R"	Tert. amine
KCN	$C{\equiv}N:^-$	RCN	Nitrile (cyanide)
AgCN	Ag-CN:	RNC (isocyanide)	Isonitrile
KNO_2	O=N—O	R—O—N=O	Alkyl nitrite
$AgNO_2$	Ag—Ö—N=O	$R—NO_2$	Nitroalkane
R'COOAg	$R'COO^-$	R'COOR	Ester
$LiAlH_4$	H	RH	Hydrocarbon
R" M^+	R"	RR'	Alkane

- cyanide and nitrite groups each offer **two nucleophilic centres**, which earns them the name **ambident nucleophiles**
- the cyanide group is a hybrid of two contributing structures, [$^-C{\equiv}N \leftrightarrow :C{=}N^-$], so it can act as a nucleophile in **two different ways** — linking through either end, continued on sheet 13

Worked example 6.e

With KCN a haloalkane yields mainly the alkyl cyanide, while AgCN leaves the isocyanide as the chief product. Explain.

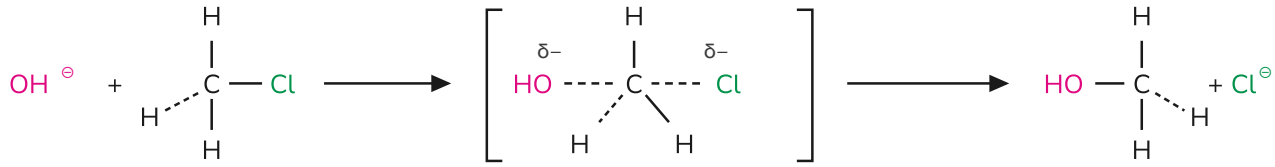
- KCN is **largely ionic**, so it puts cyanide ions into solution
- carbon and nitrogen are both placed to donate an electron pair, but attack comes **mainly through carbon**, because the **C—C bond is more stable than the C—N bond**
- AgCN is **mainly covalent**, so nitrogen is free to donate its electron pair and **isocyanide** is the main product

Ambident nucleophiles, continued

- linking through **carbon** gives alkyl cyanides, through **nitrogen** isocyanides
- the **nitrite ion** is ambident too, with two points of linkage [–O—N=O]
- through **oxygen** it gives alkyl nitrites, through **nitrogen** nitroalkanes

Mechanism · two of them

- nucleophilic substitution proceeds by **two different mechanisms**
- **(a) Substitution nucleophilic bimolecular (S_N2)** — CH_3Cl with hydroxide ion gives methanol and chloride ion, and follows **second order kinetics**: the rate depends on the concentration of **both** reactants



- in these drawings a **solid wedge** is a bond coming out of the paper, a **dashed line** a bond going down into it, and a **straight line** a bond in the plane of the paper

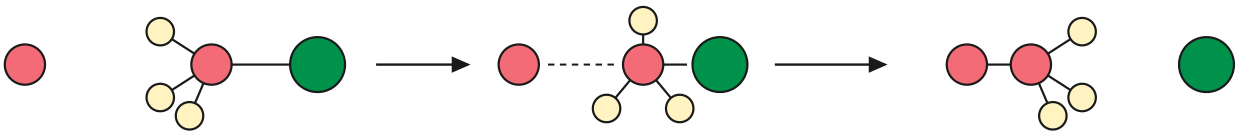


Figure 6.B Red ball represents the incoming hydroxide ion and green ball represents the outgoing halide ion

- the figure sets out a **bimolecular nucleophilic substitution (S_N2)**: the incoming nucleophile meets the alkyl halide, the **carbon–halide bond breaks** and a **new bond forms between carbon and the attacking nucleophile** — here the C–O bond between C and –OH
- the two processes run **at the same time**, in a single step — continued on sheet 14

- the S_N2 mechanism was put forward in 1937, by **Edward Davies Hughes** and **Sir Christopher Ingold**

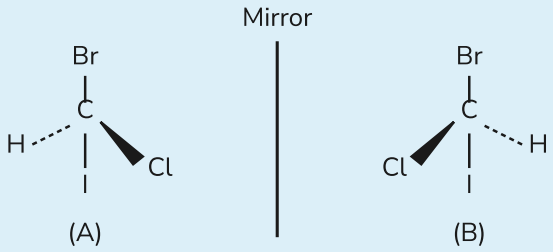
- the whole thing happens in a **single step**, and **no intermediate is formed**
- as the bond between the incoming nucleophile and carbon starts to form, the bond between **carbon and the leaving group weakens**
- the substrate's three C–H bonds then begin to **swing away from the attacking nucleophile**
- at the transition state those three C–H bonds share **one plane**, while the arriving and the departing nucleophile each hold only a **part of a bond** to carbon
- the C–H bonds keep moving the same way until the nucleophile is attached and the leaving group has gone

- the result is that the **configuration is inverted**: the carbon under attack turns over much as an **umbrella is turned inside out** in a strong wind
- the process is called **inversion of configuration**

- there carbon holds the arriving nucleophile and the departing group **at the same moment**
- a structure of that kind is **too unstable to isolate** — carbon is bound to **five atoms** at once

Configuration

- the **spatial arrangement of functional groups around a carbon** is its **configuration**

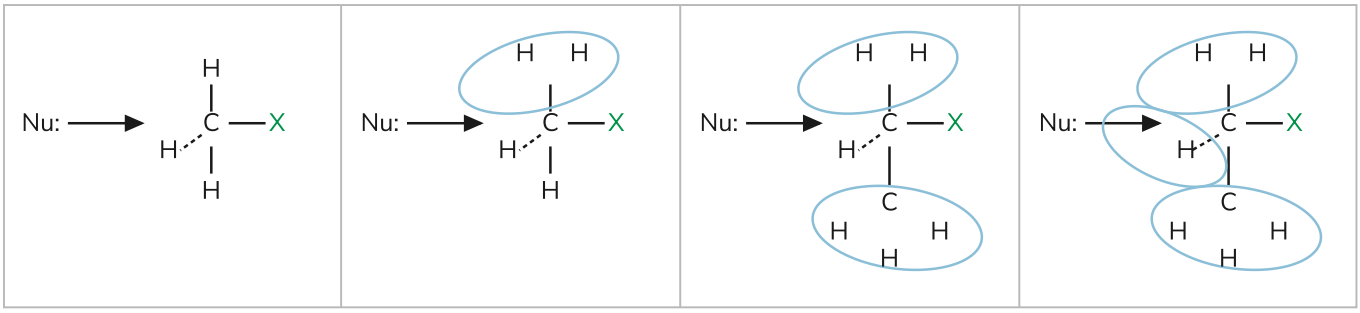


- one compound, drawn twice: the two differ in **how the groups on carbon sit in space**
- structure **(A) is the mirror image of structure (B)**, so the configuration of carbon in (A) is the mirror image of the configuration of carbon in (B)

- because the reaction needs the nucleophile to **reach the carbon carrying the leaving group**, **bulky substituents** on or near that carbon have a **dramatic inhibiting effect**
- among simple alkyl halides, **methyl halides react fastest** in S_N2 reactions — there are only three small hydrogen atoms in the way
- **tertiary halides are the least reactive**, since bulky groups hinder the approach — continued on sheet 15

- Hughes studied under Ingold, taking a D.Sc. from the University of London

- reactivity therefore falls away from the **primary halide**, through the **secondary**, to the **tertiary**



Methyl (30) | Ethyl 1° (1) | Isopropyl 2° (0.02) | *tert*-butyl 3° (0)

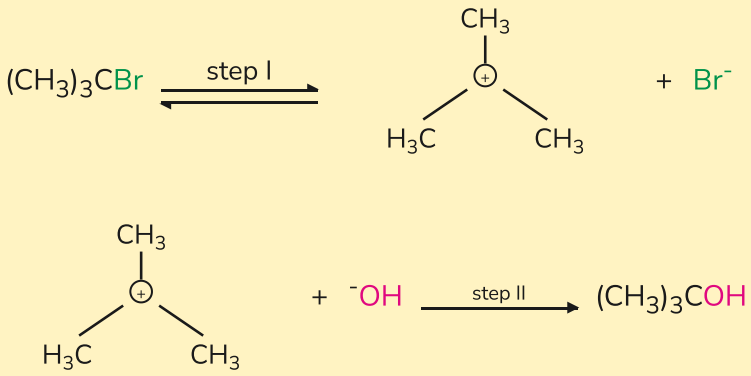
Figure 6.C Steric effects in S_N2 reaction. The relative rate of S_N2 reaction is given in parenthesis

(b) Substitution nucleophilic unimolecular (S_N1)

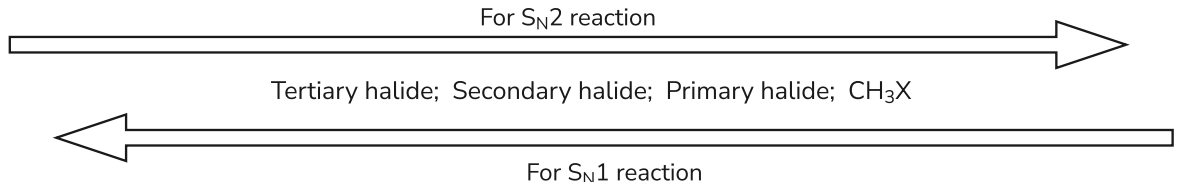
- S_N1 reactions are generally run in **polar protic solvents** — water, alcohol, acetic acid and the like
- *tert*-butyl bromide with hydroxide ion gives *tert*-butyl alcohol and follows **first order kinetics**: the rate depends on the concentration of **only one** reactant, *tert*-butyl bromide

$$\underset{\text{2-Bromo-2-methylpropane}}{(CH_3)_3CBr} + {}^-OH \longrightarrow \underset{\text{2-Methylpropan-2-ol}}{(CH_3)_3COH} + Br^-$$

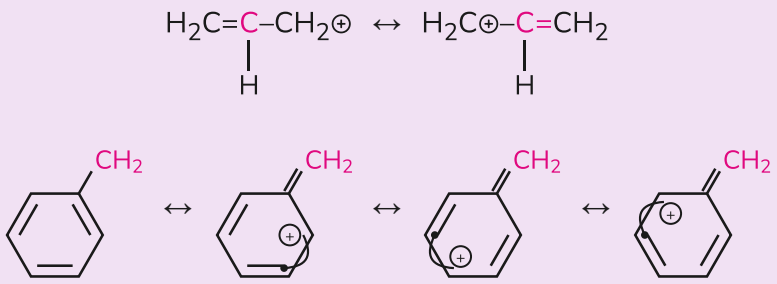
- it happens in **two steps**
- **step I** — the polarised C—Br bond is **cleaved slowly**, leaving a **carbocation** together with a bromide ion
- **step II** — the carbocation is then attacked by the nucleophile, completing the substitution



- **step I is the slowest and reversible**: it breaks the C–Br bond, and the energy for that comes from **solvation of the halide ion** with the proton of the protic solvent
- since the rate of reaction follows the slowest step, it depends **only on the concentration of alkyl halide**, not on that of hydroxide ion
- the **more stable the carbocation**, the easier it forms from the alkyl halide and the **faster the reaction**
- so **3° alkyl halides undergo S_N1 very fast**, because 3° carbocations are highly stable



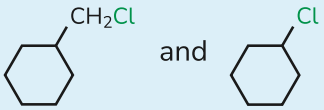
- on the same grounds, **allylic and benzylic halides react readily by S_N1**
- the carbocation formed is **stabilised through resonance** (Unit 8, Class XI)



- with the alkyl group fixed, R–X keeps **one order under either mechanism**: **R–I > R–Br > R–Cl >> R–F**

Worked example 6.f

In each pair of halogen compounds, which would undergo S_N2 reaction faster?



and ; $CH_3CH_2CH_2CH_2I$ and $CH_3CH_2CH_2CH_2Cl$

- CH_2Cl on the ring — it is a **primary halide**, so it undergoes S_N2 faster
- the **iodide** — iodine, being large, **leaves more readily**, so it goes faster once the nucleophile arrives

Worked example 6.g

Predict the order of reactivity of these compounds in S_N1 and S_N2 reactions.

- (i) the four isomeric bromobutanes
- (ii) $C_6H_5CH_2Br$, $C_6H_5CH(C_6H_5)Br$, $C_6H_5CH(CH_3)Br$, $C_6H_5C(CH_3)(C_6H_5)Br$

Worked example 6.g · the answer

(i) $CH_3CH_2CH_2CH_2Br < (CH_3)_2CHCH_2Br < CH_3CH_2CH(Br)CH_3 < (CH_3)_3CBr$ (S_N1)

$CH_3CH_2CH_2CH_2Br > (CH_3)_2CHCH_2Br > CH_3CH_2CH(Br)CH_3 > (CH_3)_3CBr$ (S_N2)

- of the two primary bromides, the carbocation from $(CH_3)_2CHCH_2Br$ is **more stable** than the one from $CH_3CH_2CH_2CH_2Br$, because of the **greater electron donating inductive effect** of the $(CH_3)_2CH–$ group, so it is more reactive in S_N1
- $CH_3CH_2CH(Br)CH_3$ is a **secondary** bromide and $(CH_3)_3CBr$ a **tertiary** one, which sets the S_N1 order above
- S_N2 reactivity runs the **reverse** way, as **steric hindrance** around the electrophilic carbon grows along that order

(ii) $C_6H_5C(CH_3)(C_6H_5)Br > C_6H_5CH(C_6H_5)Br > C_6H_5CH(CH_3)Br > C_6H_5CH_2Br$ (S_N1)

$C_6H_5C(CH_3)(C_6H_5)Br < C_6H_5CH(C_6H_5)Br < C_6H_5CH(CH_3)Br < C_6H_5CH_2Br$ (S_N2)

- of the two secondary bromides, the carbocation from $C_6H_5CH(C_6H_5)Br$ is more stable than the one from $C_6H_5CH(CH_3)Br$, because **two phenyl groups stabilise it by resonance**; so the former is the more reactive in S_N1
- a phenyl group is **bulkier than a methyl group**, so $C_6H_5CH(C_6H_5)Br$ is **less reactive** than $C_6H_5CH(CH_3)Br$ in S_N2

(c) Stereochemical aspects of nucleophilic substitution

- reading these reactions needs a few stereochemical ideas and names: **optical activity, chirality, retention, inversion, racemisation**

(i) Optical activity

- ordinary light passed through a **Nicol prism** becomes **plane polarised**; certain compounds rotate its plane, and these are **optically active**
- the angle of rotation is measured with a **polarimeter**
- rotation to the **right** (clockwise) is **dextrorotatory**, Greek for right rotating, the ***d*-form**, marked with a **(+)** before the degree of rotation
- rotation to the **left** (anticlockwise) is **laevo-rotatory**, the ***l*-form**, marked with a **(–)**
- such (+) and (–) isomers are **optical isomers**, and the phenomenon is **optical isomerism**

- William Nicol (1768-1851) built the first prism to deliver plane polarised light

(ii) Molecular asymmetry, chirality and enantiomers

- **Louis Pasteur (1848)** observed that crystals of certain compounds exist as **mirror images**, and so laid the foundation of modern stereochemistry
- he showed that aqueous solutions of both kinds of crystal rotate light **equally in magnitude** at equal concentration, but **opposite in direction**
- he held that this difference in optical activity went with the **three dimensional arrangement of atoms** in the molecules — their **configurations** — continued on sheet 18

- **J. Van't Hoff** and **C. Le Bel**, in the same year (1874) and independently, argued that the **four groups (valencies) around a central carbon are arranged tetrahedrally**
- where those four substituents **differ from one another**, the molecule's mirror image will **not lie over** the molecule
- such a carbon is an **asymmetric carbon** or **stereocentre**, and the molecule that results lacks symmetry — an **asymmetric molecule**
- that asymmetry, together with the **non-superimposability of mirror images**, is what makes such organic compounds **optically active**

- Jacobus Hendricus Van't Hoff (1852-1911) took the first Nobel Prize in Chemistry, in 1901, for work on solutions

- a **sphere, a cube, a cone** are each identical to their mirror image and can be superimposed on it
- many objects cannot: your **left and right hands** look alike, but laid in the same plane they do not coincide
- objects **non-superimposable on their mirror image**, a pair of hands among them, are **chiral**, and the property is **chirality**
- **chiral molecules are optically active**; objects and molecules that **are** superimposable are **achiral**, and optically inactive

- the same test reaches organic molecules: **build a model** of the molecule and one of its mirror image, or draw both in three dimensions and lay one over the other in the mind
- other aids help in recognising chiral molecules — one is the presence of a **single asymmetric carbon atom**
- take two simple molecules, **propan-2-ol** (Figure 6.E) and **butan-2-ol** (Figure 6.F), and their mirror images

Figure 6.D Some common examples of chiral and achiral objects

- **Chiral objects** — a left hand beside a right hand, a left footprint beside a right one: **nonsuperimposable mirror images**
- **Nonchiral objects** — a plain drinking glass beside its reflection, a circle beside its reflection: **superimposable mirror images**
- the rule the examples carry: **an object that cannot be superimposed on its mirror image is chiral**

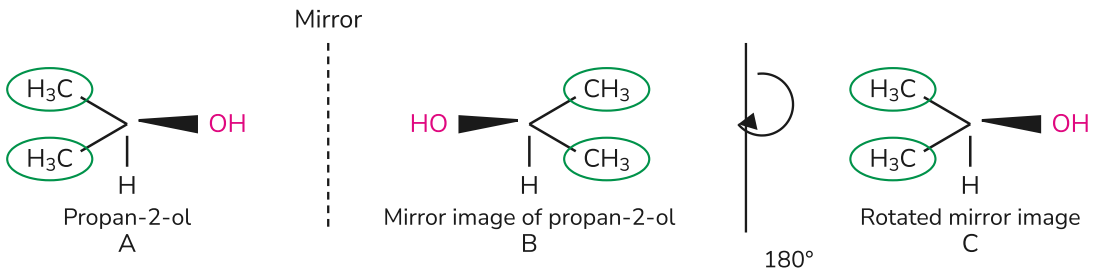


Figure 6.E B is mirror image of A; B is rotated by 180° and C is obtained; C is superimposable on A.

- propan-2-ol (A) holds **no asymmetric carbon**: the four groups on the tetrahedral carbon are **not all different**
- rotate the mirror image (B) by **180°** to reach structure (C), lay (C) over (A), and the two **overlap completely**
- so propan-2-ol is an **achiral** molecule

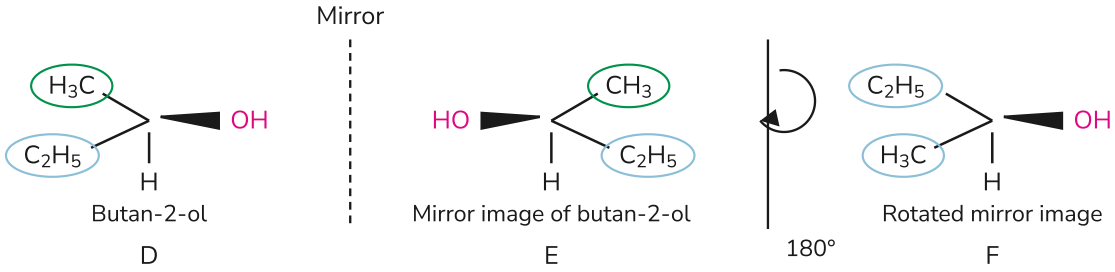


Figure 6.F E is mirror image of D; E is rotated by 180° to get F and F is non superimposable on its mirror image D.

- butan-2-ol carries **four different groups** on the tetrahedral carbon and is, as expected, **chiral**
- other common chiral molecules: **2-chlorobutane**, **2, 3-dihyroxypropanal** (OHC–CHOH–CH_2OH), **bromochloro-iodomethane** (BrClCHI), **2-bromopropanoic acid** (H_3C–CHBr–COOH)

- stereoisomers related to one another as **non-superimposable mirror images** are **enantiomers** (Figure 6.G)
- A and B in Figure 6.E, and D and E in Figure 6.F, are enantiomers
- enantiomers share **identical physical properties** — melting point, boiling point, refractive index and so on
- they differ only in the **rotation of plane polarised light**: if one is *dextro rotatory*, the other is *laevo rotatory*

Figure 6.G A chiral molecule and its mirror image

- two ball-and-stick models stand side by side, each a central carbon holding **four different groups**
- one model is the **reflection** of the other; turn either one about and the four groups **never come into the same arrangement** as its partner

- the **sign of the optical rotation** need not, however, **follow from the molecule's absolute (actual) configuration**

- a mixture holding **two enantiomers in equal proportion** has **zero optical rotation**, each isomer's rotation cancelling the other's
- such a mixture is a **racemic mixture**, or **racemic modification**, written by prefixing *dl* or (±) to the name — for example (±) butan-2-ol
- turning an enantiomer into a racemic mixture is **racemisation**

Worked example 6.h

Identify the chiral and achiral molecules in each pair below (wedge and dash representations as in Class XI).

(i) H_3C–CH(Br)(OH) and H_3C–CH(Br)(Br)

(ii) H_3C–CH(OH)–CH_2–CH_2–CH_3 and H_3C–CH_2–CH(OH)–CH_2–CH_3

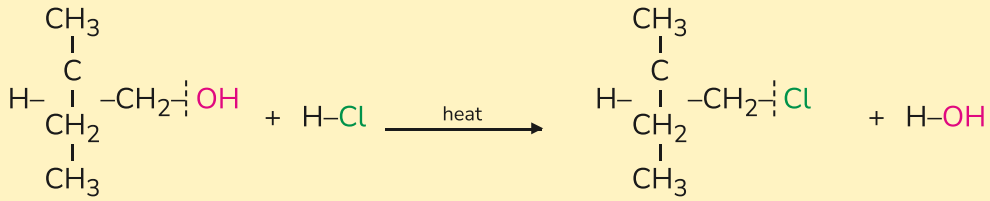
(iii) $CH_3CHBrCH_2CH_3$ and $CH_3CH_2CH_2CH_2Br$

Worked example 6.h · the answer

- (i) the first molecule is **chiral**: H, H_3C, Br and OH on one carbon are four different groups; the second, carrying **two bromines**, is achiral
- (ii) the first is **chiral**; the second, with the OH on the middle carbon of a symmetrical chain, is achiral
- (iii) $CH_3CHBrCH_2CH_3$ is **chiral**; $CH_3CH_2CH_2CH_2Br$ is achiral

(iii) Retention

- under **retention of configuration** the bonds to an asymmetric centre **keep their arrangement in space** across a chemical reaction or transformation
- in general, if **no bond to the stereocentre is broken** during a reaction, the product keeps the **same general configuration** of groups around that centre as the reactant had; the reaction then proceeds **with retention of configuration**
- take what happens when **(–)-2-methylbutan-1-ol meets hot hydrochloric acid of high concentration**

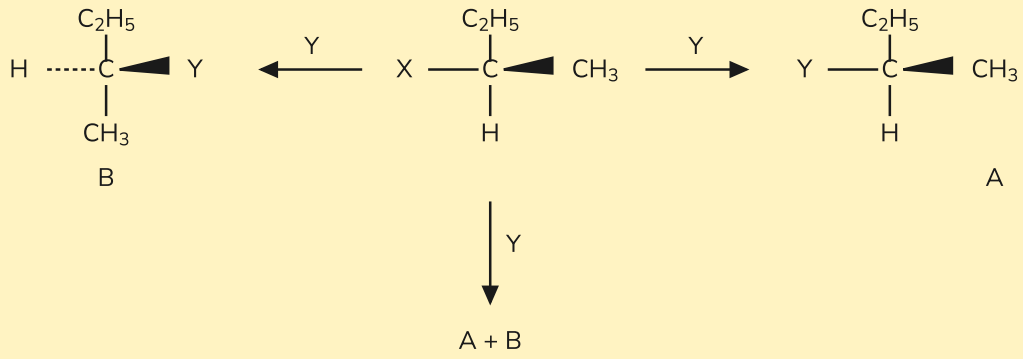


(–)-2-Methylbutan-1-ol (+)-1-Chloro-2-methylbutane

- note that the **configuration at the asymmetric centre is the same** in reactant and product, yet the **sign of optical rotation has changed**
- that is because two different compounds with the **same configuration** at an asymmetric centre may rotate light differently: one may be **dextrorotatory** (plus sign), the other **laevorotatory** (negative sign)

(iv) Inversion, retention and racemisation

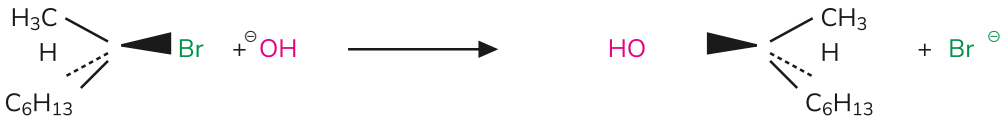
- break a bond that runs **straight to an asymmetric carbon** and the reaction can end **three ways**; let a group Y take the place of X



- where **(A) alone comes out**, the process goes by **retention of configuration**; A's configuration has merely been rotated
- where **(B) alone comes out**, it goes by **inversion of configuration**; B's configuration has been turned over

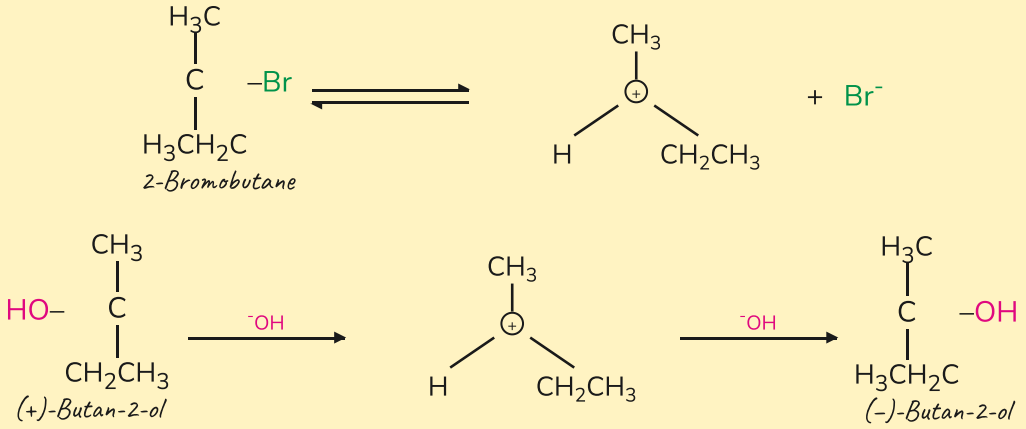
- if a **50:50 mixture of A and B** is obtained the process is **racemisation**, and the product is **optically inactive**: one isomer rotates the plane of polarised light opposite to the other
- now look afresh at S_N1 and S_N2 through **optically active alkyl halides**

- with an optically active alkyl halide, the **S_N2 product has the inverted configuration** of the reactant
- the nucleophile comes in on the **far side from the halogen**
- (–)-2-bromooctane with sodium hydroxide gives **(+)-octan-2-ol**, the –OH taking the position **opposite** to the one bromide had held



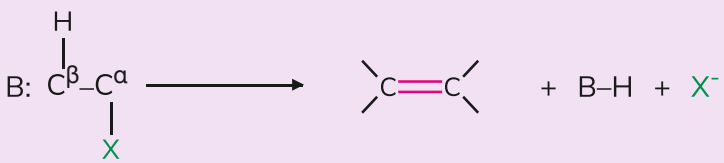
S_N2 reactions of optically active alkyl halides go with **inversion of configuration**.

- with optically active alkyl halides, **S_N1 reactions go with racemisation**
- the carbocation formed in the slow step is ***sp*^{2} hybridised and therefore planar**, which makes it **achiral**
- the nucleophile may attack from **either side of that plane**, so a **mixture of products** results: one with the same configuration, the –OH attaching where the halide ion sat, the other with the opposite configuration, the –OH attaching on the far side
- hydrolysis of optically active **2-bromobutane** shows it, giving **(±)-butan-2-ol**



2 · Elimination reactions

- heat a haloalkane that has a **β-hydrogen atom** with an **alcoholic solution of potassium hydroxide**, and hydrogen leaves the **β-carbon** while halogen leaves the **α-carbon**



B = Base ; X = Leaving group

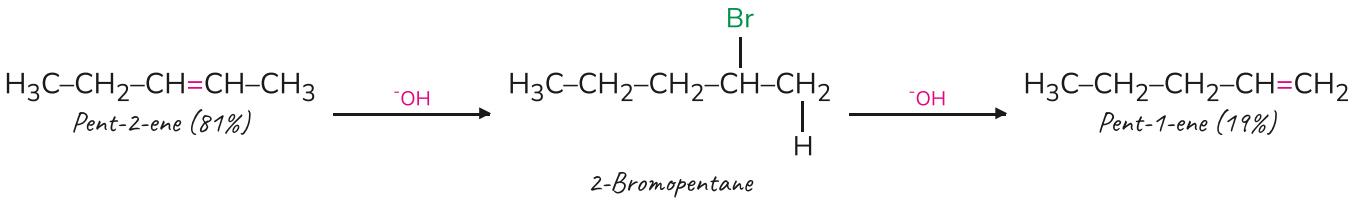
Where α and β carbon sit

- the carbon the halogen is **directly attached to** is the **α-carbon**
- the carbon **next to it** is the **β-carbon**

$$-C^{\gamma}-C^{\beta}-C^{\alpha}-X$$

- an **alkene** comes out as the product; because a **β-hydrogen atom** is the one removed, the reaction is often called **β-elimination**

- where **more than one β-hydrogen** is available, more than one alkene could form, and usually **one of them is the major product**
- the Russian chemist **Alexander Zaitsev** (also pronounced Saytzeff) first noticed the pattern and in **1875** put it as a rule: in dehydrohalogenation the favoured alkene is the one whose **doubly bonded carbons carry the greater number of alkyl groups**
- so **pent-2-ene is the major product** from 2-bromopentane



Elimination versus substitution

- a chemical reaction is a **competition**, a race won by the fastest runner: a collection of molecules does, by and large, whatever is **easiest** for it
- set a base or a nucleophile against an alkyl halide that has β-hydrogen atoms and **two routes compete** — substitution (S_N1 and S_N2), and elimination
- which route wins depends on the **nature of the alkyl halide**, the **strength and size of the base or nucleophile**, and the **reaction conditions**
- a **bulkier nucleophile** would rather act as a base and pull off a proton than come near a tetravalent carbon, for steric reasons — and the other way about for a small one
- a **primary** alkyl halide prefers S_N2; a **secondary** one takes S_N2 or elimination according to the strength of the base or nucleophile; a **tertiary** one takes S_N1 or elimination according to the stability of the carbocation or of the more substituted alkene

Elimination

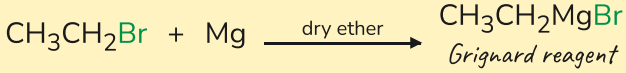
- a **bulky alkoxide base**, $(CH_3)_2CH–O^{\ominus}$, reaches past the carbon skeleton for a **hydrogen on the β-carbon**
- the C–H bond and the C–Br bond both break, the bromide leaves, and a **double bond closes** between the two carbons

Substitution

- a **small nucleophile**, $CH_3CH_2–O^{:-}$, goes straight at the **carbon carrying the bromine**
- the new C–O bond forms as the **C–Br bond breaks**, and the bromide is displaced

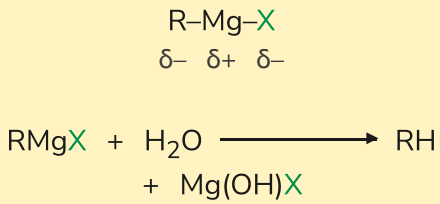
3 · Reaction with metals

- set most organic **chlorides, bromides and iodides** against certain metals and the products hold a **carbon–metal bond**: these are the **organo-metallic compounds**
- an important class of them was discovered by **Victor Grignard in 1900**: the **alkyl magnesium halide RMgX**, known as a **Grignard Reagent**
- these come from haloalkanes with **magnesium metal in dry ether**



- **Victor Grignard** began academic life oddly for a chemist: he took a **maths degree**
- when he did move to chemistry it was not to the mathematical side of physical chemistry but to **organic chemistry**
- hunting an efficient catalyst for **methylation**, he saw that **Zn in diethyl ether** had served, and wondered whether **Mg with ether** might do the same
- Grignard reagents were **first reported in 1900**, and he used the work for his **doctoral thesis in 1901**
- in **1910** he took a professorship at the University of Nancy, and in **1912** the **Nobel prize for Chemistry**, shared with **Paul Sabatier**, who had advanced nickel catalysed hydrogenation

- the **carbon–magnesium bond of a Grignard reagent is covalent, yet strongly polar**: carbon draws electrons off the electropositive magnesium
- the bond from magnesium to halogen is **ionic for all practical purposes**
- a Grignard reagent is **highly reactive**: **anything that can give up a proton** converts it to a hydrocarbon, and water, alcohols and amines are acidic enough to do so
- so **every trace of moisture must be kept out**, which is why the reaction is run in **dry ether**; seen the other way, this is one method of **converting halides to hydrocarbons**



Wurtz reaction

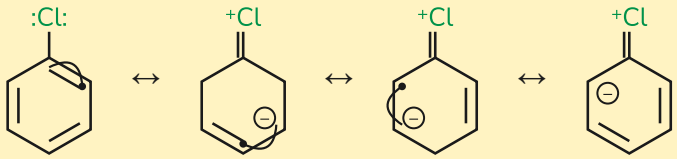
- put alkyl halides to **sodium in dry ether** and the hydrocarbon that results holds **twice as many carbon atoms** as the halide did



A.8 Chemical reactions of haloarenes

1 · Nucleophilic substitution

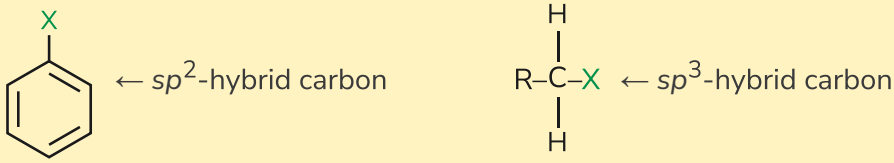
- aryl halides show **extremely little reactivity** in nucleophilic substitution, for the reasons below
- **(i) Resonance effect** — in a haloarene the **electron pairs on the halogen are in conjugation with the π-electrons of the ring**, so these resonating structures are possible



- the **C—Cl bond takes on partial double bond character** through that resonance
- as a result the bond is **harder to break** in a haloarene than in a haloalkane, and haloarenes are therefore **less reactive towards nucleophilic substitution**

(ii) The carbon of the C–X bond is hybridised differently

- in a **haloalkane** the carbon attached to halogen is ***sp*^{3} hybridised**; in a **haloarene** it is ***sp*^{2} hybridised**



- the *sp*^{2} carbon has **greater *s*-character**, so it is **more electronegative** and holds the electron pair of the C—X bond **more tightly** than the *sp*^{3} carbon of a haloalkane, which has less *s*-character
- hence the **C—Cl bond length is 177 pm in a haloalkane but 169 pm in a haloarene**
- a **shorter bond is harder to break** than a longer one, so in nucleophilic substitution a haloarene lags behind a haloalkane

(iii) Instability of the phenyl cation

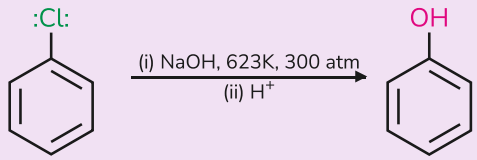
- with haloarenes the **phenyl cation** that self-ionisation would produce is **not stabilised by resonance**, so the **S_N1 mechanism is ruled out**

(iv) Repulsion

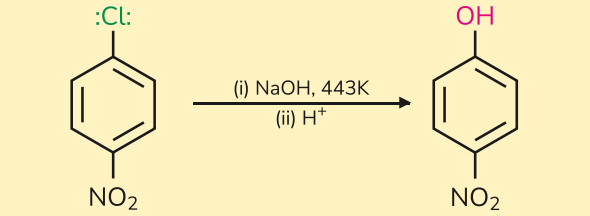
- an **electron rich nucleophile** is unlikely to approach an **electron rich arene**, the two repelling one another

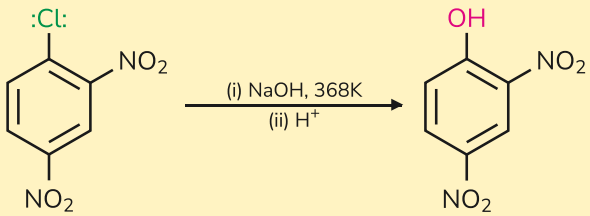
Replacement by hydroxyl group

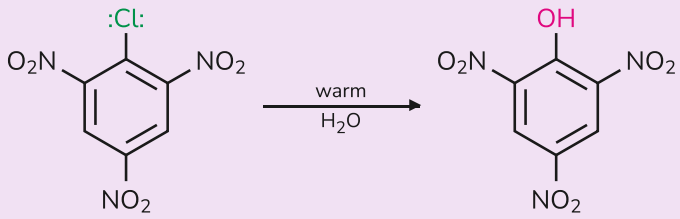
- heating **chlorobenzene in aqueous sodium hydroxide** at **623 K** under **300 atmospheres** turns it into **phenol**



- an **electron withdrawing group** (–NO_2) at the ***ortho* and *para* positions raises the reactivity** of haloarenes

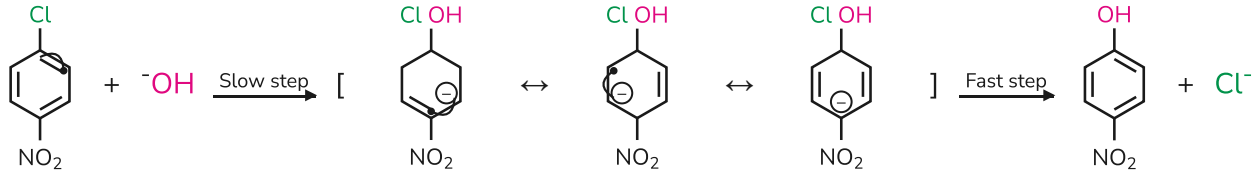




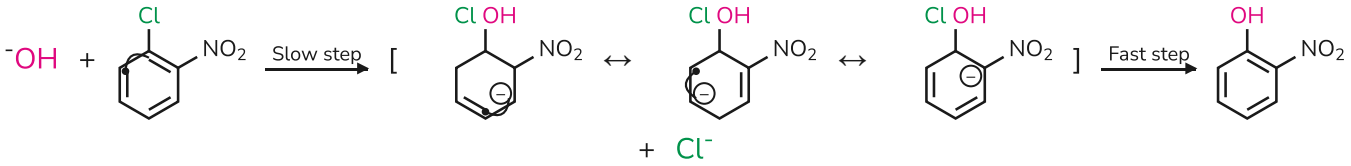


- the effect is **pronounced when the (-NO_2) group sits at the *ortho* and *para* positions**
- an electron withdrawing group at the ***meta* position has no effect** on the reactivity of a haloarene
- the mechanism below: a **slow step** builds a resonance-stabilised carbanion, a **fast step** expels chloride

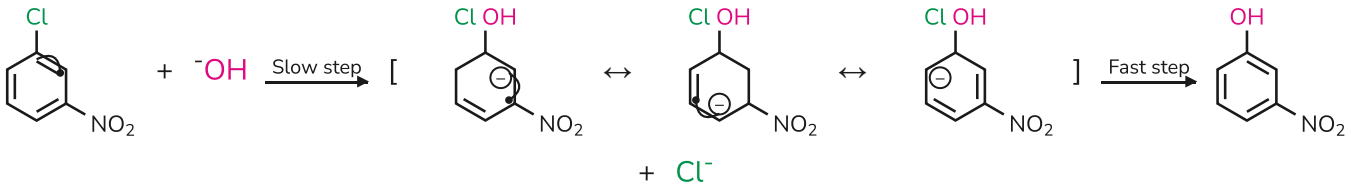
p-isomer



o-isomer



m-isomer

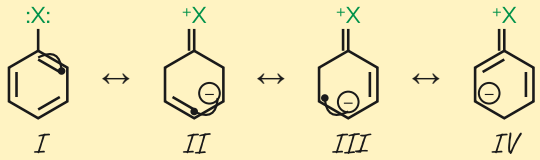


Why the NO_2 group works only at ortho and para

- a nitro group at the ***ortho* or *para* position withdraws electron density from the benzene ring**, which makes it easier for the nucleophile to attack the haloarene
- resonance then **stabilises the carbanion** produced
- counted from the halogen substituent, the negative charge falls at the *ortho* and *para* positions, where the **–NO_2 group holds it steady**
- with ***meta*-nitrobenzene none of the resonating structures carries the negative charge on the carbon bearing the –NO_2 group**
- a nitro group at the *meta* position therefore **leaves the negative charge unsupported**, and **reactivity is unaffected**

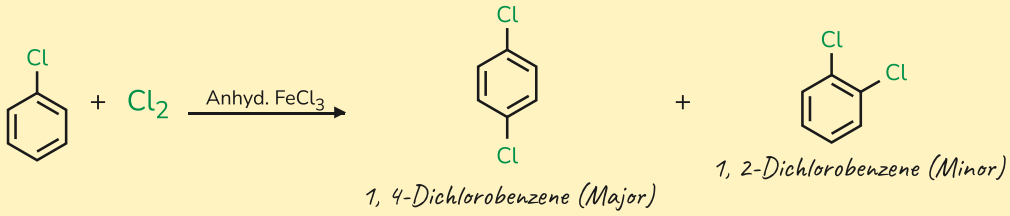
2 · Electrophilic substitution reactions

- the **electrophilic reactions a benzene ring usually enters** are open to haloarenes — halogenation, nitration, sulphonation and Friedel-Crafts reactions
- the halogen atom is **slightly deactivating** but also ***o*, *p*-directing**, so further substitution comes at the ***ortho* and *para* positions** with respect to it
- that directing influence follows from the resonating structures of a halobenzene

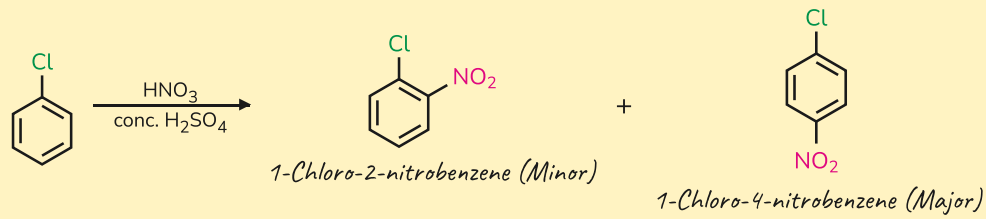


- through resonance the **electron density rises more at the *ortho* and *para* positions** than at the *meta* ones
- the halogen also has a **–I effect**, so it tends to **withdraw electrons** from the ring
- the ring is therefore **somewhat deactivated** compared with benzene, and electrophilic substitution in haloarenes runs **slowly and needs more drastic conditions**

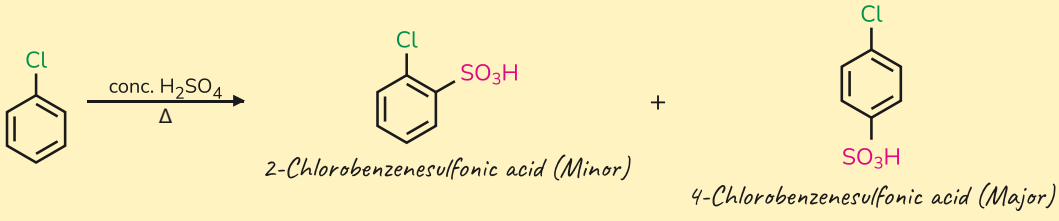
(i) Halogenation



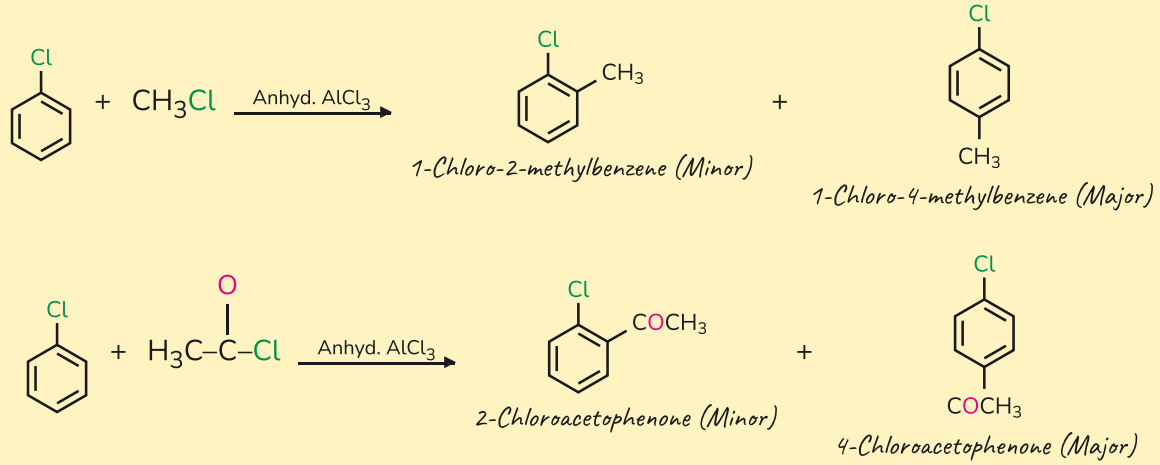
(ii) Nitration



(iii) Sulphonation



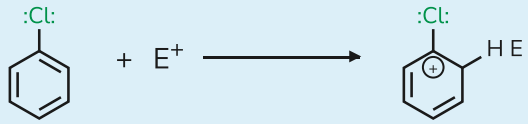
(iv) Friedel-Crafts reaction



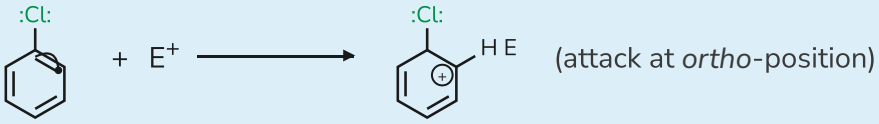
Worked example 6.i

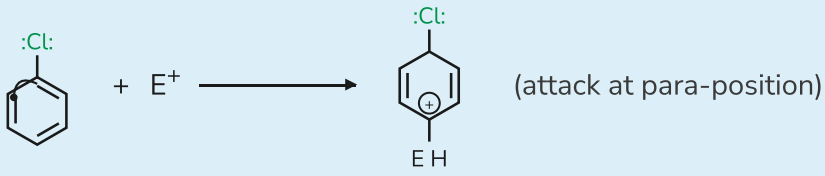
Chlorine pulls electrons away, and yet in aromatic electrophilic substitution it sends the incoming group to the *ortho* and *para* positions. Why?

- by the **inductive effect chlorine withdraws electrons**; by **resonance it releases them**
- through the inductive effect it **destabilises the intermediate carbocation** formed during the substitution



the inductive effect destabilises the intermediate carbocation





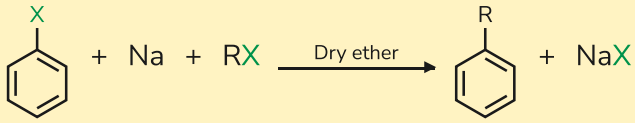
the resonance effect stabilises the intermediate carbocation

- resonance has the halogen **propping the carbocation up**, and it does so **most at the *ortho* and *para* positions**
- of the two the **inductive effect is the stronger**, so electrons are **withdrawn on balance and the ring is deactivated**
- the resonance effect **opposes** the inductive effect for attack at the *ortho* and *para* positions, making the **deactivation less** there
- so **reactivity is controlled by the stronger inductive effect**, while **orientation is controlled by the resonance effect**

3 · Reaction with metals

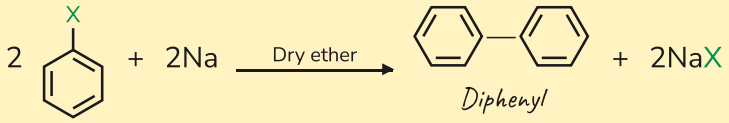
Wurtz-Fittig reaction

- put **sodium in dry ether** to a **mixture of alkyl halide with aryl halide** and an **alkylarene** results; this is the **Wurtz-Fittig reaction**



Fittig reaction

- aryl halides alone give analogous compounds with **sodium in dry ether**, **two aryl groups joining together**; this is the **Fittig reaction**



Intext questions

Q7 Which alkyl halide of each pair would you expect to react more rapidly by an S_N2 mechanism? Explain your answer.

(i) $CH_3CH_2CH_2CH_2Br$ or $CH_3CH_2CH(Br)CH_3$

(ii) $CH_3CH_2CH(Br)CH_3$ or $H_3C–C(CH_3)_2–Br$

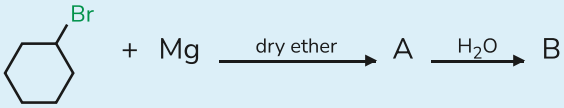
(iii) $CH_3CH(CH_3)CH_2Br$ or $CH_3CH_2CH(CH_3)CH_2Br$

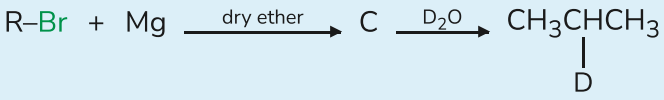
Q8 Which member of each pair below reacts faster by S_N1?

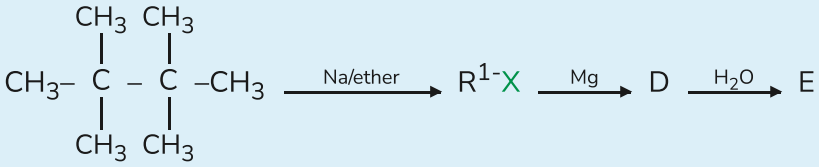
(i) $(CH_3)_3C–Cl$ and $CH_3CH_2CH(Cl)CH_3$

(ii) $CH_3CH_2CH_2CH_2CH(Cl)CH_3$ and $CH_3CH_2CH_2CH_2CH_2CH_2Cl$

Q9 Identify A, B, C, D, E, R and R^1 in the following.







B.1 Polyhalogen compounds

- carbon compounds holding **more than one halogen atom** are usually called **polyhalogen compounds**
- many of them are useful in **industry and agriculture**; several are described here

Dichloromethane (methylene chloride)

- widely used as a **solvent**, a **paint remover**, a **propellant in aerosols** and a **process solvent in drug manufacture**
- also a **metal cleaning and finishing** solvent
- it **harms the human central nervous system**
- **lower levels in air** — slightly impaired hearing and vision
- **higher levels in air** — dizziness and nausea, with tingling and numbness reaching the fingers and toes
- **direct skin contact** brings intense burning and mild redness; in **the eyes** it can burn the cornea

Trichloromethane (chloroform)

- it dissolves **fats, alkaloids and iodine**, among other substances
- its **major use today** is in producing the **freon refrigerant R-22**
- once a **general anaesthetic** in surgery, since replaced by less toxic, safer ones such as ether
- inhaling its vapours **depresses the central nervous system**; about **900 parts per million** of air for a short time causes dizziness, fatigue and headache
- **chronic exposure** may damage the **kidneys**, and the **liver**, which metabolises chloroform to phosgene; skin held in it raises sores on some people
- it is **slowly oxidised by air in the presence of light** to the extremely poisonous gas **carbonyl chloride**, also called **phosgene**
- so it is stored in **closed dark coloured bottles filled completely**, to keep air out

Triiodomethane (iodoform)

- used earlier as an **antiseptic**, though the antiseptic action comes from the **liberation of free iodine**, not from iodoform itself
- its **objectionable smell** has seen it replaced by other iodine-containing formulations

Tetrachloromethane (carbon tetrachloride)

- produced in large quantities for making **refrigerants** and **propellants for aerosol cans**
- also a **feedstock** for chlorofluorocarbons and other chemicals, for **pharmaceutical manufacturing**, and for general solvent use
- until the **mid 1960s** widely used as a **cleaning fluid** — a degreasing agent in industry, a spot remover and fire extinguisher in the home
- there is some evidence that exposure **causes liver cancer in humans**
- commonest of all are **dizziness and light headedness, nausea, vomiting**, and these can leave nerve cells permanently damaged; severe cases pass quickly into **stupor, coma, unconsciousness or death**
- exposure may set the **heart beating irregularly, or stop it**; on contact the chemical can irritate the eyes
- let into the air it **climbs to the atmosphere, where it depletes the ozone layer** — continued on sheet 30

- a thinner ozone layer is held to **let more ultraviolet reach us**, and with it more **skin cancer**, more **disease and disorder of the eye**, and possible **damage to the immune system**

Freons

- **freon** is the collective name for the **chlorofluorocarbons derived from methane and ethane**
- they are gases that liquefy easily, and are **extremely stable, unreactive, non-toxic and non-corrosive**
- industry's commonest is **Freon 12 (CCl_2F_2)**, made from **tetrachloromethane by the Swarts reaction**
- the usual purposes are **aerosol propellants, refrigeration and air conditioning**
- world production stood at roughly **2 billion pounds a year** by **1974**
- nearly all of it, refrigeration's share included, finds the atmosphere in the end and **diffuses unchanged into the stratosphere**, where it can set off **radical chain reactions that upset the natural ozone balance**

p,p′-Dichlorodiphenyltrichloroethane (DDT)

- the **first chlorinated organic insecticide**, originally prepared in **1873**
- not until **1939** did **Paul Muller** of Geigy Pharmaceuticals in Switzerland discover its effectiveness as an insecticide; for that he took the **1948 Nobel Prize in Medicine and Physiology**
- use rose enormously worldwide after the Second World War, chiefly for its effect on the **mosquito that spreads malaria** and the **lice that carry typhus**
- problems appeared in the **late 1940s**: many insect species **developed resistance**, and DDT proved **highly toxic to fish**
- its **chemical stability and fat solubility** compounded the problem: animals metabolise it only slowly, so it is **deposited and stored in fatty tissues** and **builds up over time** where ingestion continues at a steady rate
- its use was **banned in the United States in 1973**, though it is still used in some other parts of the world

Summary

- an alkyl or aryl halide counts as a **mono, di or polyhalogen** compound (tri-, tetra- and so on) according to how many halogen atoms it holds
- halogen atoms are **more electronegative than carbon**, so the carbon–halogen bond of an alkyl halide is **polarised**: the carbon carries a partial positive charge and the halogen a partial negative one
- the routes to an alkyl halide are **halogenating an alkane by the free radical path**, **adding a halogen acid to an alkene**, and **swapping an alcohol's –OH for halogen** with a phosphorus halide, thionyl chloride or a halogen acid — continued on sheet 31

- aryl halides are prepared by **electrophilic substitution of arenes**; **fluorides and iodides are best made by halogen exchange**
- strong **dipole-dipole and van der Waals** attraction lifts the boiling point of an organohalogen compound **above that of the matching hydrocarbon**; water takes up only a little of it, an organic solvent all of it
- the **polarity of the carbon–halogen bond** is what opens alkyl halides to **nucleophilic substitution**, to **elimination**, and to **metal atoms**, which build organometallic compounds
- nucleophilic substitutions are sorted into **S_N1 and S_N2** by their **kinetics**; chirality matters greatly in reading those mechanisms — **S_N2 of a chiral alkyl halide inverts the configuration**, while **S_N1 racemises**
- polyhalogen compounds — dichloromethane, chloroform, iodoform, carbon tetrachloride, freon, DDT — have many **industrial applications**, but several **decompose only with difficulty**, some **deplete the ozone layer**, and they are proving **environmental hazards**

Exercises

E1 Give each halide below its IUPAC name, and say whether it is an alkyl, allyl or benzyl halide (primary, secondary or tertiary), a vinyl halide or an aryl halide: (i) $(CH_3)_2CHCH(Cl)CH_3$; (ii) $CH_3CH_2CH(CH_3)CH(C_2H_5)Cl$; (iii) $CH_3CH_2C(CH_3)_2CH_2I$; (iv) $(CH_3)_3CCH_2CH(Br)C_6H_5$; (v) $CH_3CH(CH_3)CH(Br)CH_3$; (vi) $CH_3C(C_2H_5)_2CH_2Br$; (vii) $CH_3C(Cl)(C_2H_5)CH_2CH_3$; (viii) $CH_3CH{=}C(Cl)CH_2CH(CH_3)_2$; (ix) $CH_3CH{=}CHC(Br)(CH_3)_2$; (x) *p*-$ClC_6H_4CH_2CH(CH_3)_2$; (xi) *m*-$ClCH_2C_6H_4CH_2C(CH_3)_3$; (xii) *o*-Br-$C_6H_4CH(CH_3)CH_2CH_3$

E2 Give the IUPAC names of: (i) $CH_3CH(Cl)CH(Br)CH_3$; (ii) $CHF_2CBrClF$; (iii) $ClCH_2C{\equiv}CCH_2Br$; (iv) $(CCl_3)_3CCl$; (v) $CH_3C(p\text{-}ClC_6H_4)_2CH(Br)CH_3$; (vi) $(CH_3)_3CCH{=}CClC_6H_4I$-*p*

E3 Draw the structure of each of these: (i) 2-Chloro-3-methylpentane; (ii) *p*-Bromochlorobenzene; (iii) 1-Chloro-4-ethylcyclohexane; (iv) 2-(2-Chlorophenyl)-1-iodooctane; (v) 2-Bromobutane; (vi) 4-*tert*-Butyl-3-iodoheptane; (vii) 1-Bromo-4-*sec*-butyl-2-methylbenzene; (viii) 1,4-Dibromobut-2-ene

E4 Whose dipole moment is the largest — (i) CH_2Cl_2, (ii) $CHCl_3$ or (iii) CCl_4?

E5 In the dark a hydrocarbon C_5H_{10} leaves chlorine alone; in bright sunlight it yields one monochloro compound only, C_5H_9Cl. Name the hydrocarbon.

E6 Set out the isomers of C_4H_9Br.

E7 Give equations that make 1-iodobutane from (i) 1-butanol, (ii) 1-chlorobutane, (iii) but-1-ene.

E8 Say what an ambident nucleophile is, and illustrate it.

E9 Against ^{-}OH in an S_N2 reaction, which member of each pair is the faster? (i) CH_3Br or CH_3I; (ii) $(CH_3)_3CCl$ or CH_3Cl

E10 Dehydrohalogenate each halide below with sodium ethoxide in ethanol. Predict every alkene formed and pick out the major one. (i) 1-Bromo-1-methylcyclohexane; (ii) 2-Chloro-2-methylbutane; (iii) 2,2,3-Trimethyl-3-bromopentane

E11 Carry out each conversion below. (i) Ethanol to but-1-yne; (ii) Ethane to bromoethene; (iii) Propene to 1-nitropropane; (iv) Toluene to benzyl alcohol; (v) Propene to propyne; (vi) Ethanol to ethyl fluoride; (vii) Bromomethane to propanone; (viii) But-1-ene to but-2-ene; (ix) 1-Chlorobutane to *n*-octane; (x) Benzene to biphenyl

E12 Account for each of these. (i) Chlorobenzene has a smaller dipole moment than cyclohexyl chloride. (ii) Polar as they are, alkyl halides will not mix with water. (iii) A Grignard reagent must be made in the absence of water.

E13 What is each of freon 12, DDT, carbon tetrachloride and iodoform used for?

E14 Draw the major organic product of each reaction below: (i) $CH_3CH_2CH_2Cl + NaI$; (ii) $(CH_3)_3CBr + KOH$; (iii) $CH_3CH(Br)CH_2CH_3 + NaOH$; (iv) $CH_3CH_2Br + KCN$; (v) $C_6H_5ONa + C_2H_5Cl$; (vi) $CH_3CH_2CH_2OH + SOCl_2$; (vii) $CH_3CH_2CH = CH_2 + HBr$; (viii) $CH_3CH = C(CH_3)_2 + HBr$

E15 Write the mechanism of the reaction nBuBr + KCN ⟶ nBuCN

E16 Put each set in order of how readily it takes S_N2 displacement: (i) 2-Bromo-2-methylbutane, 1-Bromopentane, 2-Bromopentane; (ii) 1-Bromo-3-methylbutane, 2-Bromo-2-methylbutane, 2-Bromo-3-methylbutane; (iii) 1-Bromobutane, 1-Bromo-2,2-dimethylpropane, 1-Bromo-2-methylbutane, 1-Bromo-3-methylbutane

E17 Which of $C_6H_5CH_2Cl$ and $C_6H_5CHClC_6H_5$ does aqueous KOH hydrolyse the more easily?

E18 *p*-Dichlorobenzene has a higher m.p. than the *o*- and *m*-isomers. Discuss.

E19 Say how each conversion may be made. (i) Propene to propan-1-ol; (ii) Ethanol to but-1-yne; (iii) 1-Bromopropane to 2-bromopropane

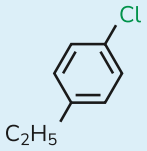
E19 (continued) (iv) Toluene to benzyl alcohol; (v) Benzene to 4-bromonitrobenzene; (vi) Benzyl alcohol to 2-phenylethanoic acid; (vii) Ethanol to propanenitrile; (viii) Aniline to chlorobenzene; (ix) 2-Chlorobutane to 3, 4-dimethylhexane; (x) 2-Methyl-1-propene to 2-chloro-2-methylpropane; (xi) Ethyl chloride to propanoic acid; (xii) But-1-ene to *n*-butyliodide; (xiii) 2-Chloropropane to 1-propanol; (xiv) Isopropyl alcohol to iodoform; (xv) Chlorobenzene to *p*-nitrophenol; (xvi) 2-Bromopropane to 1-bromopropane; (xvii) Chloroethane to butane; (xviii) Benzene to diphenyl; (xix) *tert*-Butyl bromide to isobutyl bromide; (xx) Aniline to phenylisocyanide

E20 Alkyl chlorides give alcohols with aqueous KOH, but alkenes as the major product with alcoholic KOH. Explain.

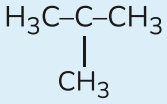
E21 A primary alkyl halide C_4H_9Br (a) gave compound (b) with alcoholic KOH. Compound (b) with HBr gave (c), an isomer of (a). Sodium metal turned (a) into compound (d), C_8H_{18}, unlike the product sodium gives with *n*-butyl bromide. Draw the structural formula of (a) and set out equations for every reaction.

E22 Say what results in each case. (i) Alcoholic KOH acting on *n*-butyl chloride. (ii) Mg acting on bromobenzene in dry ether. (iii) Chlorobenzene put to hydrolysis. (iv) Aqueous KOH acting on ethyl chloride. (v) Sodium acting on methyl bromide in dry ether. (vi) KCN acting on methyl chloride.

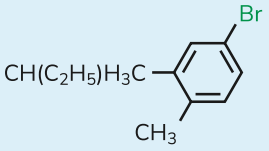
Answers to some intext questions

Q1 (i) $CH_3CH_2CH(CH_3)CHClCH_3$ (ii) 

(iv) $BrCH_2CH = CHCH_2Br$

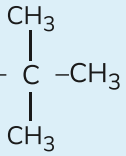
(iii) $CH_3CH_2CH_2CHCH(I)CH_2CH_3$ 

on the carbon marked above

(v) 

- **Q2** H_2SO_4 has no place beside KI in making an alkyl iodide from an alcohol: it converts the KI to HI, and then oxidises that HI to I_2
- **Q3** (i) $ClCH_2CH_2CH_2Cl$ (ii) $ClCH_2CHClCH_3$ (iii) $Cl_2CHCH_2CH_3$ (iv) $CH_3CCl_2CH_3$

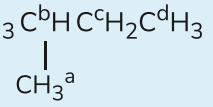
Q4

(i) 

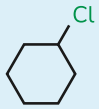
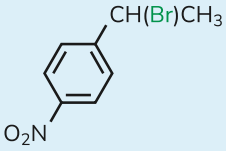
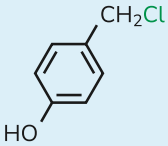
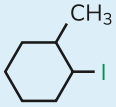
all the hydrogen atoms are equivalent, so replacing any of them gives the same product

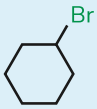
(ii) $C^aH_3C^bH_2C^cH_2C^bH_2C^aH_3$

equivalent hydrogens group as a, b and c; replacing them gives the same product

(iii)  grouped as a, b, c and d, so **four** isomeric products are possible

Q5

(i)  (ii)  (iii)  (iv)  (v)

CH_3CH_2I (vi) 

Q6

- (i) Chloromethane, Bromomethane, Dibromomethane, Bromoform — the **boiling point climbs with the molecular mass**
- (ii) Isopropylchloride, 1-Chloropropane, 1-Chlorobutane — **branching** puts isopropylchloride below 1-chloropropane

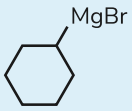
Q7

- (i) $CH_3CH_2CH_2CH_2Br$ — a **primary halide**, so there is no steric hindrance
- (ii) $CH_3CH_2CH(Br)CH_3$ — a **secondary halide reacts faster than a tertiary one**
- (iii) $CH_3CH(CH_3)CH_2CH_2Br$ — a methyl group **closer to the halide** raises the steric hindrance and lowers the rate

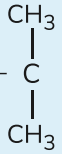
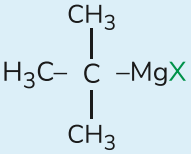
Q8

- (i) $(CH_3)_3C–Cl$ — a **tertiary halide reacts faster than a secondary one**, the *tert*-carbocation being more stable
- (ii) $CH_3CH_2CH_2CH_2CH(Cl)CH_3$ — because a **secondary carbocation is more stable than a primary one**

Q9

A =  B = 

C = $RMgBr$ R = CH_3CHCH_3

R 1 =  D = 

E = 